

HEART DISEASE PREDICTION USING LOGISTIC REGRESSION

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INTRODUCTION

Cardiovascular diseases are a leading cause of mortality. Logistic regression can help predict the likelihood of heart disease based on clinical data, enabling timely interventions. This project aims to develop a logistic regression model for accurate heart disease prediction.

OBJECTIVES

- Develop a logistic regression model for predicting heart disease.
- Analyze significant predictors of heart disease.
- Evaluate the model using accuracy, precision, recall, and AUC-ROC metrics.

DATASET

The UCI Heart Disease Dataset will be used, featuring patient records with attributes such as age, sex, cholesterol, and blood pressure. The target variable indicates the presence or absence of heart disease.

METHODOLOGY

- Preprocess the dataset by handling missing values, encoding variables, and scaling data.
- Conduct exploratory data analysis to identify trends and relationships.
- Train and optimize the logistic regression model, evaluating performance with relevant metrics.

TOOLS AND TECHNOLOGIES

- Python
- Libraries: pandas, numpy, scikit-learn, matplotlib
- Development Environment: Jupyter Notebook

EXPECTED OUTCOMES

- A reliable predictive model for heart disease.
- Insights into critical factors contributing to heart disease risk.
- A comprehensive report summarizing the methodology and findings.

REFERENCES

UCI Machine Learning Repository Heart Disease Data Set